

The Momentum Hinge

One rate equation, three mechanisms of momentum transfer

Working rule: Force measures the rate at which momentum changes or crosses a chosen system boundary. Begin with $\mathbf{F}_{\text{ext}} = d\mathbf{p}_{\text{system}}/dt$, state the system, and then add the collision condition, momentum flux or molecular model that makes the force calculable.

Name: _____

Class: _____

Companion task to *The Placeholder Force*

1. Collision: impulse and internal force

A 0.20 kg trolley moving at 4.0 m s^{-1} to the right collides with a stationary 0.30 kg trolley. They lock together. The contact lasts 0.050 s and external horizontal impulse is negligible.

A. Represent before and after

Draw both trolleys before, during and after impact. Choose a positive direction. Show all velocities and draw the contact-force pair during the collision. Mark two possible systems: trolley 1 alone, and both trolleys together.

B. Conserve the system momentum

Use momentum conservation to find their common final velocity. Identify the extra physical statement that closes the problem. Explain why momentum conservation alone would not determine two separate final velocities.

C. Apply the hinge to each trolley

For each trolley calculate Δp and the average contact force,

$$\mathbf{F}_{\text{av}} = \frac{\Delta \mathbf{p}}{\Delta t}.$$

Compare the two forces. Then apply $\mathbf{F}_{\text{ext}} = d\mathbf{p}/dt$ to the combined two-trolley system during the collision.

D. Interrogate the equation

How can each trolley have a large momentum change while the system momentum remains constant? What role is played by Newton’s third law? Which conclusion depends on the short collision time?

2. Rocket thrust: momentum crossing a boundary

At one instant a vertical rocket has mass 1200 kg. It ejects gas at 30 kg s^{-1} with speed 600 m s^{-1} relative to the rocket. The nozzle pressure matches the atmosphere. Take $g = 9.8\text{ m s}^{-2}$ and neglect drag.

A. Choose the system boundary

Sketch the rocket, nozzle and exhaust. Draw one boundary around the rocket alone and a second around rocket plus recently expelled gas. Mark the direction in which mass and momentum cross the first boundary.

B. Turn momentum flux into thrust

Over a short time Δt , show that expelled mass Δm carries relative momentum of magnitude $u_e \Delta m$. Hence use

$$T = \dot{m} u_e$$

to calculate the thrust. Find the thrust impulse delivered in 0.20 s.

C. Join thrust to the rocket’s motion

Calculate the rocket’s weight, instantaneous net force and acceleration. Write a signed momentum balance showing why the rocket gains upward momentum as the exhaust carries downward momentum away.

D. Interrogate variable mass

Why can $T = \dot{m} u_e$ not be obtained by treating the rocket as a closed fixed-mass object? What extra term would appear if nozzle exit pressure differed from atmospheric pressure?

3. Gas pressure: molecular momentum flux

A cube of side $L = 0.20\text{ m}$ contains $N = 2.0 \times 10^{23}$ identical molecules, each of mass $4.65 \times 10^{-26}\text{ kg}$. Their rms speed is 500 m s^{-1} . Assume elastic wall collisions and an isotropic gas.

A. One molecule, one wall

For a molecule with velocity component v_x towards a wall, show that an elastic collision changes its momentum by magnitude $2mv_x$. The time between successive collisions with the same wall is $2L/|v_x|$. Use $F = \Delta p/\Delta t$ to obtain its average force contribution.

B. From molecules to pressure

Sum over all molecules and use $V = AL$ to show

$$P = \frac{F}{A} = \frac{Nm\langle v_x^2 \rangle}{V}.$$

Use isotropy, $\langle v_x^2 \rangle = \frac{1}{3}\langle v^2 \rangle$, to obtain $P = Nm c_{\text{rms}}^2/(3V)$.

C. Calculate and connect to energy

Calculate the gas pressure. Then show that

$$PV = \frac{1}{3}Nm c_{\text{rms}}^2 = \frac{2}{3}E_{\text{k,total}}.$$

Explain how this provides a microscopic route towards the ideal-gas law.

D. Interrogate pressure

Is pressure a substance stored in the gas, a force, or momentum flux per unit area? Why is random molecular motion compatible with a steady macroscopic pressure?

Synthesis: force gives the rate; the system model identifies the transfer

1. Complete the comparison.

Case	Momentum system	Extra closure or model	What produces the force?
Collision			
Rocket			
Gas pressure			

2. Develop the central claim.

Respond to:

“Force is a rate of momentum transfer, but the boundary decides how the transfer is seen.”

Use all three cases. Include the terms *external impulse*, *internal force*, *system boundary*, *momentum flux*, *variable mass*, *elastic collision*, and *isotropy*. Explain why the same equation $\mathbf{F} = d\mathbf{p}/dt$ can describe a violent contact force, smooth rocket thrust and steady gas pressure.

Teacher guide and indicative answers

The Momentum Hinge

1. Collision: impulse and system momentum

Common final velocity. The “lock together” condition supplies $v_{1f} = v_{2f} = v_f$. Conservation gives

$$0.20(4.0) + 0.30(0) = 0.50v_f,$$

so

$$v_f = 1.6 \text{ m s}^{-1}.$$

Conservation alone provides one scalar equation; without a collision condition it cannot determine two unknown final velocities.

Momentum changes.

$$\Delta p_1 = 0.20(1.6 - 4.0) = -0.48 \text{ N s},$$

$$\Delta p_2 = 0.30(1.6 - 0) = +0.48 \text{ N s}.$$

Therefore

$$F_{1,\text{av}} = -9.6 \text{ N}, \quad F_{2,\text{av}} = +9.6 \text{ N}.$$

The forces are equal and opposite, consistent with Newton’s third law. Their impulses redistribute momentum internally.

For the combined system,

$$\Delta p_{\text{system}} = 0, \quad F_{\text{ext,av}} \approx 0.$$

Large internal forces can therefore coexist with no change in total momentum.

Role of collision time. The impulse is fixed by Δp ; a shorter contact time means a larger average force. The assumption of negligible external impulse is what permits system momentum conservation during the brief event.

Diagnostic misconception. “Momentum is conserved for each trolley.” It is conserved for the isolated two-trolley system; each trolley exchanges momentum with the other.

2. Rocket thrust: open-system momentum

With pressure matched at the nozzle, thrust is the exhaust momentum flux:

$$T = \dot{m}u_e = 30(600) = 1.80 \times 10^4 \text{ N}.$$

Over 0.20 s,

$$J_T = T\Delta t = 3600 \text{ N s}.$$

Equivalently, 6.0 kg of gas is expelled and carries relative momentum $6.0(600) = 3600 \text{ kg m s}^{-1}$ downward.

The rocket’s weight is

$$W = mg = 1200(9.8) = 1.176 \times 10^4 \text{ N}.$$

Thus

$$F_{\text{net}} = 18000 - 11760 = 6240 \text{ N}$$

and the instantaneous acceleration is

$$a = \frac{6240}{1200} = 5.2 \text{ m s}^{-2}$$

upward.

System interpretation. For rocket plus exhaust, momentum is redistributed: exhaust gains downward momentum and the rocket gains upward momentum. For the rocket alone, mass and momentum cross the boundary, so a fixed-mass equation applied without a flux term is incomplete. In one-dimensional form,

$$m \frac{dv}{dt} = T + F_{\text{external}}, \quad T = u_e(-\dot{m}_{\text{rocket}}).$$

If exit and ambient pressures differ, the fuller expression is

$$T = \dot{m}u_e + (p_e - p_a)A_e.$$

Diagnostic misconception. Exhaust “pushes on the air.” Rocket thrust persists in vacuum because it follows from momentum carried by the exhaust; ambient pressure only adds or subtracts the pressure-thrust term.

3. Gas pressure: kinetic-theory route

For one elastic wall collision,

$$|\Delta p_x| = 2m|v_x|.$$

The same molecule returns to that wall after $\Delta t = 2L/|v_x|$, giving

$$F_i = \frac{|\Delta p_x|}{\Delta t} = \frac{mv_x^2}{L}.$$

Summing over molecules,

$$F = \frac{m}{L} \sum_i v_{x,i}^2 = \frac{Nm\langle v_x^2 \rangle}{L}.$$

Since $V = AL$,

$$P = \frac{F}{A} = \frac{Nm\langle v_x^2 \rangle}{V}.$$

For an isotropic gas,

$$\langle v_x^2 \rangle = \frac{1}{3} \langle v^2 \rangle = \frac{1}{3} c_{\text{rms}}^2,$$

so

$$P = \frac{Nmc_{\text{rms}}^2}{3V}.$$

Here $V = (0.20)^3 = 8.0 \times 10^{-3} \text{ m}^3$, hence

$$P = \frac{(2.0 \times 10^{23})(4.65 \times 10^{-26})(500)^2}{3(8.0 \times 10^{-3})} \approx 9.7 \times 10^4 \text{ Pa}.$$

Also,

$$E_{\text{k,total}} = \frac{1}{2}Nmc_{\text{rms}}^2, \quad PV = \frac{2}{3}E_{\text{k,total}}.$$

Together with the thermal result $E_{\text{k,total}} = \frac{3}{2}NkT$, this yields $PV = NkT$.

Conceptual answer. Pressure is the normal momentum flux per unit area. Random microscopic impulses average to a stable macroscopic value because enormous numbers of uncorrelated collisions occur each second.

Synthesis and classroom use

Indicative synthesis.

All three cases use force as a rate of momentum transfer, but the system boundary changes the bookkeeping. In the collision, equal and opposite internal impulses redistribute momentum while the total remains fixed because external impulse is negligible. In the rocket, momentum is carried through an open boundary by expelled mass; thrust is a momentum flux, supplemented by a pressure term when nozzle and ambient pressures differ. In the gas, repeated molecular impulses deliver momentum to a wall, and pressure is that momentum-transfer rate per unit area. The rate equation supplies the hinge; the collision condition, exhaust model and kinetic assumptions identify the mechanism.

Discussion prompts.

- For which system are the collision forces internal, and for which are they external?
- Why does a rocket require an open-system momentum balance?
- What exactly crosses the rocket boundary: mass, momentum, or both?
- Why is gas pressure naturally described as momentum flux rather than stored force?
- Which assumptions produce the factor 1/3 in kinetic theory?

Suggested timing: 10 minutes collision, 10 minutes rocket, 12 minutes kinetic theory, 10 minutes synthesis and whole-class comparison.